

ENTERPRISE JAVA

Mini Project Report

College Fest Event Management & Ticketing System

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1. Background of the Project

College festivals are among the most anticipated events in educational institutions. They include cultural programs, technical competitions, workshops, seminars, sports events, coding contests, guest lectures, and exhibitions. These events often attract a large number of participants, making event management a complex task.

In many colleges, the event registration process is still handled manually through paper forms or spreadsheets. Attendance is marked using printed lists, and certificates are created individually after the event. These traditional methods consume considerable time, increase paperwork, and are prone to human errors such as duplicate registrations, incorrect attendance records, and loss of participant information. Managing seat availability and preparing reports also becomes difficult when multiple events are conducted simultaneously.

The proposed College Fest Event Management & Ticketing System is designed to overcome these challenges by providing a centralized web-based platform. It automates the complete event lifecycle — from creating events and registering participants to generating QR-code-based tickets, verifying attendance through QR scanning, and issuing digital certificates. The system also provides an administrative dashboard that displays registrations, seat occupancy, revenue, and attendance statistics.

This project aims to improve efficiency, reduce manual work, enhance data security, and provide a smooth experience for students, organizers, and administrators.

Enterprise Java technologies such as JSP, Servlets, MySQL, and Apache Tomcat are used in this project to develop a secure, scalable, and efficient web-based event management system with dynamic content and reliable database management.

2. Objectives

To develop a centralized web-based platform for managing college fest events, registrations, ticketing, attendance, and certificate generation.

To automate the event registration process by allowing students to browse available events and register online, reducing manual paperwork and administrative effort.

To provide secure role-based access for students, organizers, and administrators through authentication and authorization mechanisms.

To generate unique QR code-based e-tickets for every registered participant, ensuring fast, secure, and paperless event entry.

To simplify attendance management by verifying QR codes at the venue and automatically recording participant attendance.

To automate digital certificate generation for eligible participants, eliminating manual certificate preparation and distribution.

To enable organizers to efficiently create, update, and manage events, including scheduling, venue allocation, seat capacity, and registration fees.

To provide an administrative dashboard that offers real-time insights into registrations, attendance, seat occupancy, and revenue through reports and statistics.

To improve data accuracy, security, and reliability by storing event-related information in a centralized database and preventing duplicate registrations or unauthorized access.

To build a scalable and maintainable Enterprise Java application using JSP, Servlets, MySQL, and Apache Tomcat that can support future enhancements such as online payments, mobile integration, and notifications.

3.. Functional Requirements

Functional requirements describe the features and services that the system must provide to its users. The system is divided into eight core modules, described below.

Module 1: User Registration and Login

This module allows students, organizers, and administrators to create accounts and access the system securely. Authentication ensures that users can only perform actions according to their roles.

Functions

- New user registration
- Secure login
- Password validation
- Forgot password functionality
- Profile management
- Logout

Inputs	Outputs
Name	User details stored
Email ID	Email verified
Password	Password encrypted and validated
Role (Student/Organizer/Admin)	User account created and dashboard displayed

Module 2: Event Management

This module is accessible to organizers and administrators. It allows them to create and manage college events efficiently.

Functions

- Create new events
- Edit event details
- Delete events
- Set event date and time
- Specify venue
- Define registration fee
- Set maximum seat capacity
- Upload event description and banner

Inputs	Outputs
Event name	Event successfully created
Venue	Event details saved successfully
Date	Event added to the event list
Time	Event timing displayed
Registration fee	Registration fee updated
Maximum Seat Capacity	Available seats updated
Description	Event details published

Module 3: Student Registration for Events

Students can browse all available events and register for the ones they wish to attend.

Functions

- View available events
- Search events by category
- Register for an event
- Cancel registration
- Check seat availability
- View registration history

System Validation

- Prevent duplicate registrations
- Prevent registration after seat limit is reached
- Validate user login before registration

Outputs

- Registration confirmation
- Seat availability update
- Registration status

Module 4: QR Code Ticket Generation

Once registration is completed successfully, the system automatically generates a unique QR code for the participant.

Functions

- Generate unique QR code
- Create e-ticket
- Download ticket
- Store ticket details in database

Ticket Contains

- Student Name
- Event Name
- Registration ID
- QR Code
- Event Date
- Venue

Outputs

- Downloadable QR Ticket (PDF)

Module 5: QR Code Verification

At the venue entrance, organizers scan the participant's QR code to verify authenticity.

Functions

- Scan QR code
- Verify registration details
- Check duplicate entry
- Mark attendance automatically

Outputs

- Entry Allowed
- Invalid Ticket
- Already Scanned Message

Module 6: Attendance Management

Attendance is recorded automatically after successful QR verification.

Functions

- Mark attendance
- View attendance records
- Export attendance reports

Outputs

- Attendance list
- Participant status

- Attendance report

Module 7: Certificate Generation

Students who attend the event receive digital certificates automatically.

Functions

- Generate PDF certificates
- Add student details
- Add event details
- Add organizer signature
- Download certificate

Outputs

- PDF Certificate
- Certificate history

Module 8: Admin Dashboard

The administrator controls the entire system through a dashboard.

Functions

- Manage users
- Manage organizers
- Manage events
- View registrations
- View attendance
- Track revenue
- Generate reports

Dashboard Displays

- Total Events
- Total Students
- Total Registrations
- Seats Filled
- Revenue Generated
- Attendance Percentage

4.. Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

Security

Security is essential because the system stores user information and event records. The system should:

- Provide secure login authentication.

- Encrypt user passwords.
- Allow role-based access control.
- Prevent unauthorized access.
- Validate QR tickets before entry.
- Protect database records through backups.

Performance

The system should respond quickly even when many users access it simultaneously. Performance requirements include:

- Fast page loading.
- QR code generation within a few seconds.
- Quick event registration.
- Instant attendance verification.
- Efficient database operations.
- Support for multiple concurrent users.

Reliability

The system should work continuously without failures. Requirements:

- Accurate attendance records.
- Correct registration details.
- Reliable certificate generation.
- Database recovery after failures.

Availability

The system should be available whenever users need it. Requirements:

- 24×7 accessibility.
- Minimal downtime.
- Backup server support.

Usability

The interface should be easy to understand. Requirements:

- Simple navigation
- Attractive dashboard
- Responsive design
- User-friendly forms

Scalability

The application should be capable of handling future growth. It should support:

- Thousands of students
- Multiple events
- Multiple organizers
- Future mobile application integration

Maintainability

The software should be easy to update. Requirements:

- Modular coding
- Well-documented code
- Easy bug fixing
- Easy feature addition

5. Design of the Project

This section presents the design of the College Fest Event Management & Ticketing System using UML and structured-analysis diagrams: an Entity-Relationship (ER) diagram, a Sequence diagram covering the core registration-to-certificate flow, and Data Flow Diagrams (DFD) up to Level 2.

4.1 Entity-Relationship (ER) Diagram

The ER diagram below models the core entities of the system — User, Event, Registration, Ticket, Attendance, and Certificate — and the relationships between them.

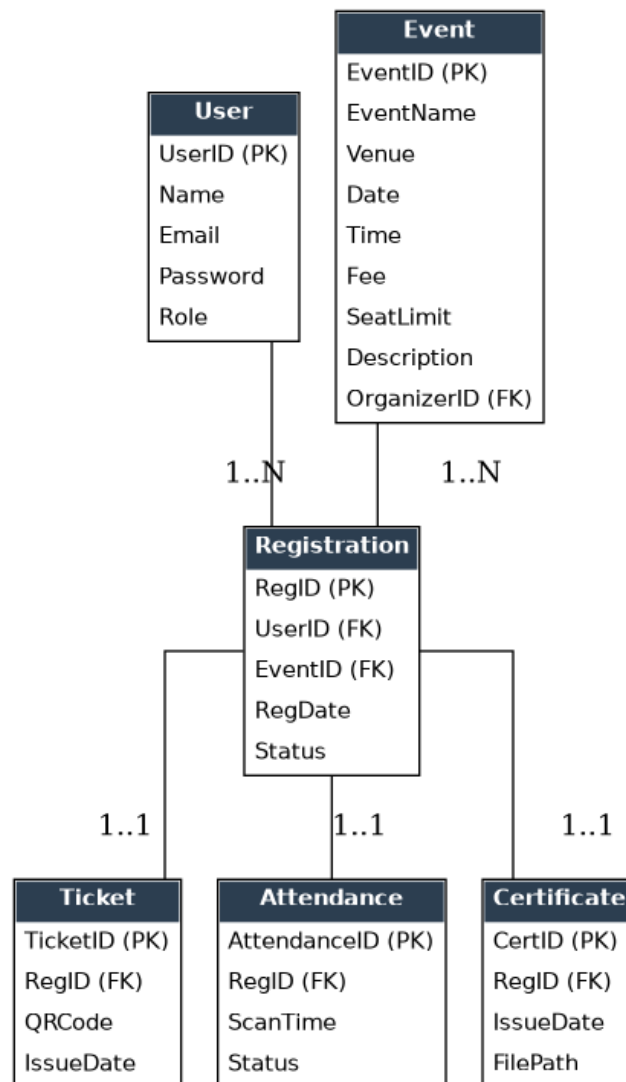


Figure 4.1: Entity-Relationship Diagram

4.2 Sequence Diagram

The sequence diagram illustrates the interaction between a Student, the System, the Database, an Organizer, and the Certificate Module — from login and event registration through QR ticket issuance, attendance verification at the venue, and certificate generation.

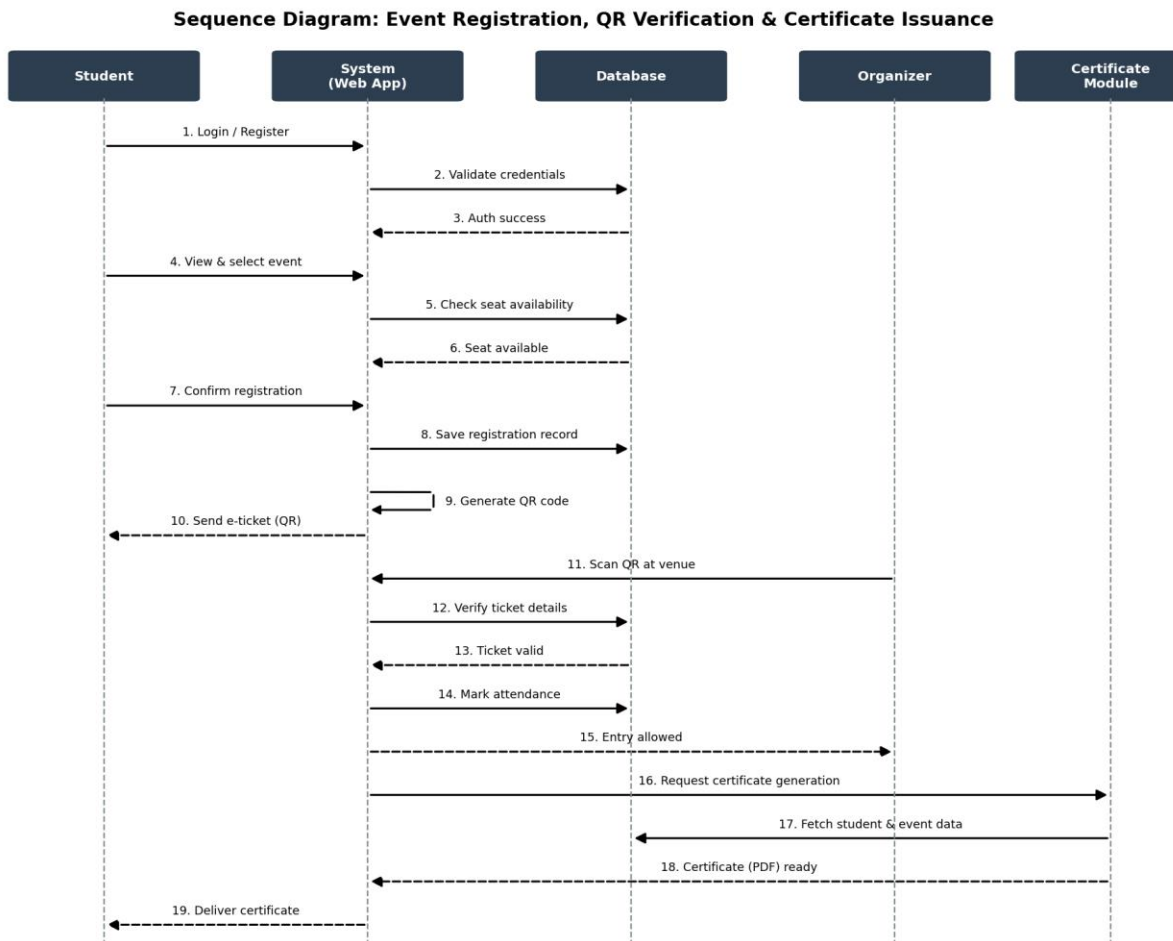


Figure 4.2: Sequence Diagram — Registration, QR Verification & Certificate Issuance

4.3 Data Flow Diagram (DFD)

Level 0 — Context Diagram

The context diagram shows the system as a single process interacting with its three external entities: Student, Organizer, and Admin.

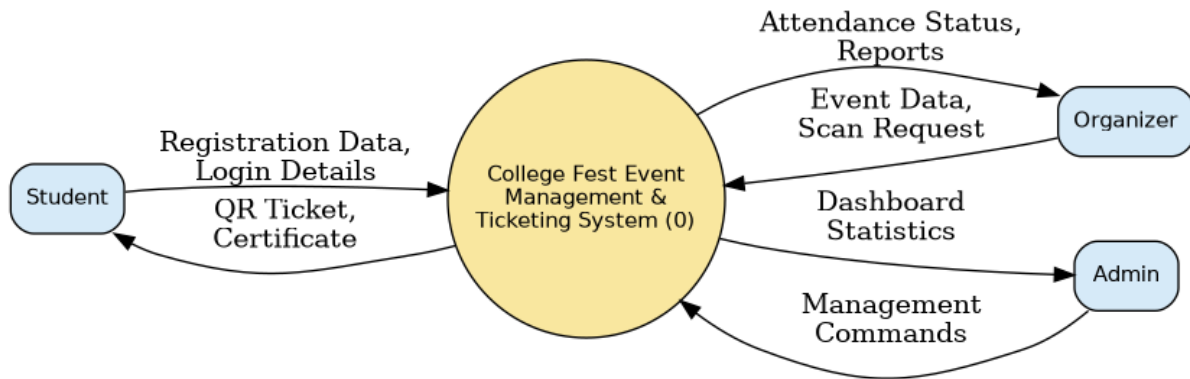


Figure 4.3: DFD Level 0 (Context Diagram)

Level 1

Level 1 decomposes the system into its eight major processes, each interacting with the relevant data stores (Users, Events, Registrations, Tickets, Attendance, Certificates).

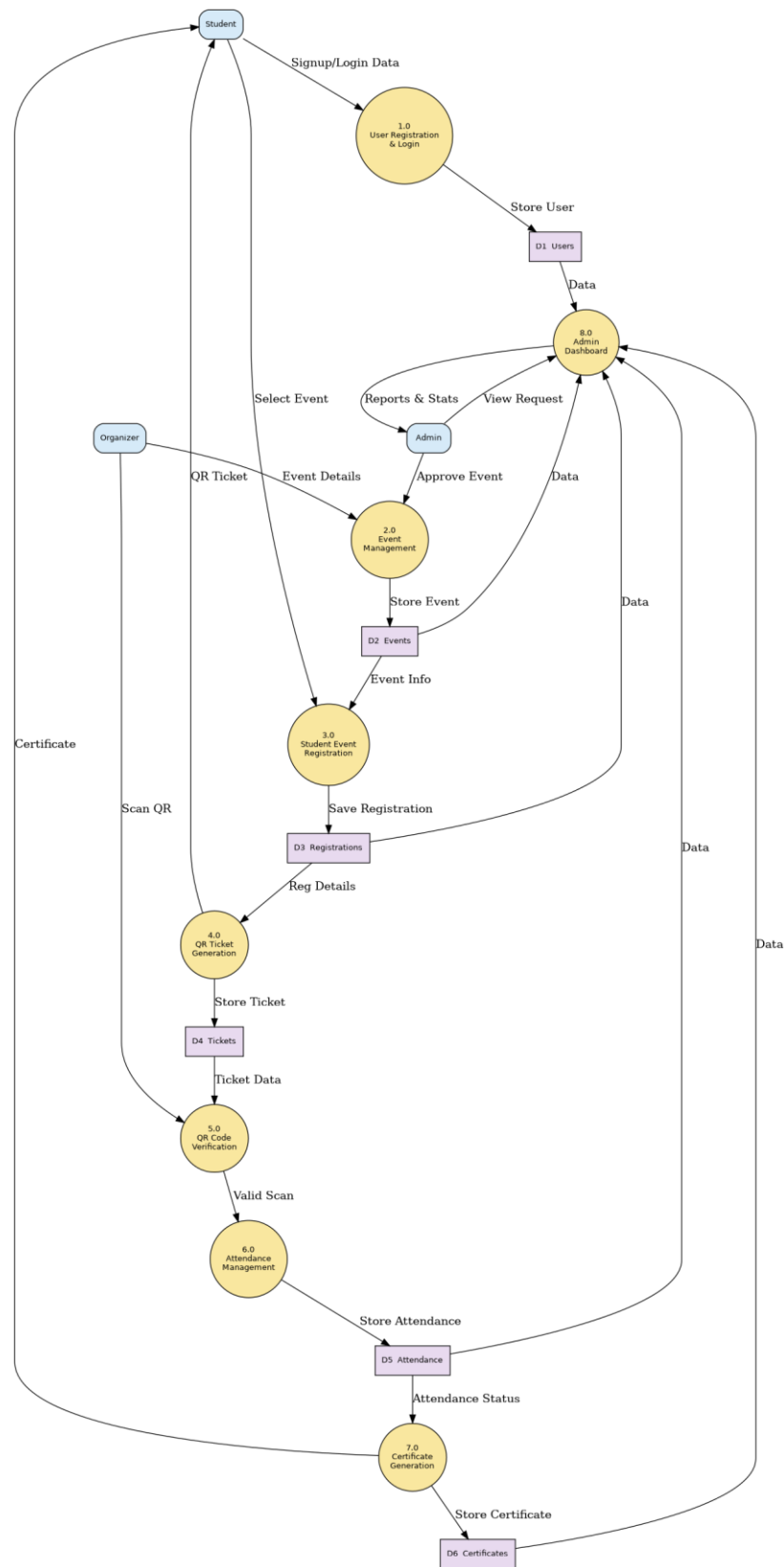


Figure 4.4: DFD Level 1

Level 2

Level 2 further decomposes Process 3.0 (Student Event Registration) and Process 4.0 (QR Ticket Generation) into their constituent sub-processes, showing how registration validation and ticket creation occur in detail.

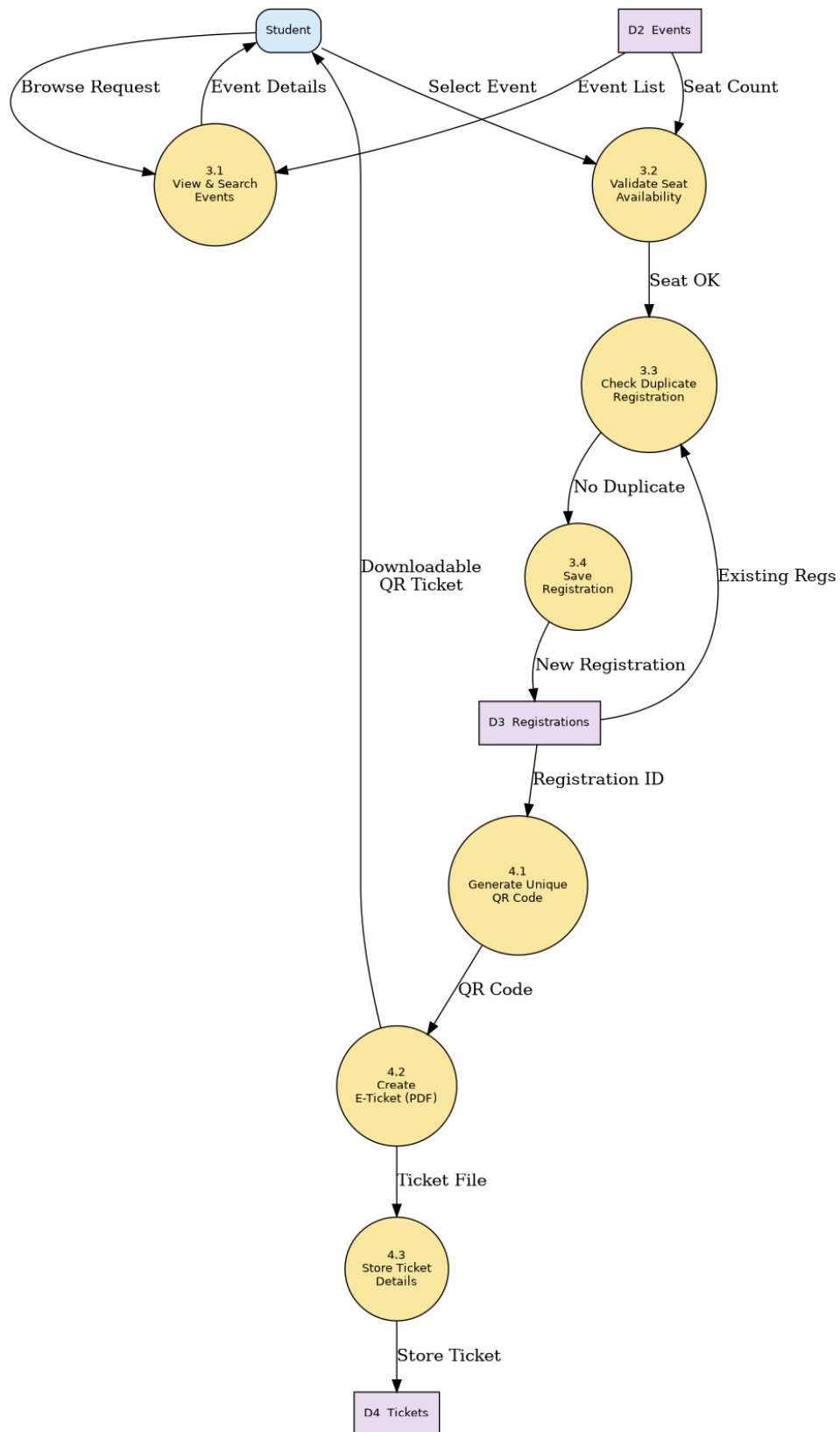


Figure 4.5: DFD Level 2 (Decomposition of Registration & Ticket Generation)